

1. What are the minimum and maximum numbers of elements in a heap of height h ?
2. Show that in any subtree of a min-heap, the root of the subtree contains the smallest value occurring anywhere in that subtree.
3. Where in a min-heap might the largest element reside, assuming that all elements are distinct?
4. Recall the following procedure to build a max-heap from an array.

```
BUILD-MAX-HEAP(A,n)
1 A.heap-size=n
2 for i=n/2 to 1
3   MAX-HEAPIFY(A,i)
```

In order to analyze the running time of this procedure, we used the following facts.

- The height of a max-heap on n elements is $\lfloor \log_2 n \rfloor$.
- The running time of $\text{MAX-HEAPIFY}(A, i)$ is $\mathcal{O}(h_i)$ where h_i denotes the height of the node at index i .

Then we concluded that the running time of $\text{BUILD-MAX-HEAP}(A, n)$ is $\mathcal{O}(n \log n)$. Let us now perform a better analysis. First show that there are $\lceil \frac{n}{2^{h+1}} \rceil$ nodes of height h (hint: you may use induction on h). Then using this result, conclude that the running time of $\text{BUILD-MAX-HEAP}(A, n)$ is $\mathcal{O}(n)$.

5. Another way to build a heap is by repeatedly calling MAX-HEAP-INSERT to insert the elements into the heap. Consider the procedure $\text{BUILD-MAX-HEAP-ALT}$ given below. It assumes that the objects being inserted are just the heap elements.

```
BUILD-MAX-HEAP-ALT(A,n)
1 A.heap-size=1
2 for i=2 to n
3   MAX-HEAP-INSERT(A,A[i],n)
```

- (a) Do the procedures BUILD-MAX-HEAP and $\text{BUILD-MAX-HEAP-ALT}$ always create the same heap when run on the same input array? Prove that they do, or provide a counterexample.
- (b) Show that in the worst case, $\text{BUILD-MAX-HEAP-ALT}$ requires $\Theta(n \log n)$ time to build an n -element heap.